Asynchronous Backward Value Iteration (ABVI)

- ABVI is a version of BVI in which states are not updated in full synchrnonized sweeps
- Instead, inbdividual states are updated *selectively* – often according to priority – while cost-to-go information propagates backward from the goal.
- Useful for understanding D* Lite.

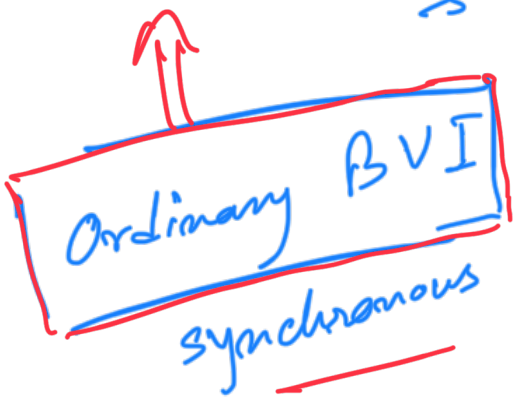
Standard BVI:

$V(s)$: opt cost from state s to goal $s_g \in X_G$

$$V(s) = \min_{s' \in Succ(s)} \left[c(s, s') + V(s') \right] ; \; V(s_g) = 0$$

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$$V_{i+1}(s) = \min_{s'} \left[c(s, s') + V_i(s') \right]$$



$\Rightarrow$ ↳ Performs a sweep for all nodes in the graph (or for all states in state space X).

Conceptually:

Iteration 1: update ALL states

" 2: " ALL "

" 3: " ALL "

until convergence

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For ABVI, we update only one state [as needed]

$$V(s_i) \leftarrow \min_{s' \in Succ(s)} \left[c(s_i, s') + V(s') \right]$$

$V(s_i)$ ↓ specific state

and immediately use the new value $V(s_i)$ when updating another state.

Suppose, $S \to A \to B \to G_2$

Initially, $V(G) = 0$, $V(S) = V(A) = V(B) = \infty$

$B \to A \to S$

First, $V(B) = 1 + V(G) = 1$

using $V(B)$, update $V(A) = 1 + V(B) = 2$

using $V(A)$, update $V(S) = 1 + V(A) = 3$

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How does D* Lite and ABVI relate ?

Thank you